

Customer Shopping Behavior Analysis

1. Project Overview

The **OLA Data Analyst Project** focuses on analyzing ride-booking data from [OLA](#) using **SQL** and **Power BI** to generate meaningful business insights. The project helps in understanding customer behavior, ride trends, booking performance, cancellations, payment preferences, revenue generation, and customer/driver satisfaction.

Using SQL queries and Power BI dashboards, the project provides interactive visualizations and analytical reports that support data-driven decision-making. Key insights include ride volume trends, booking status analysis, vehicle-wise performance, revenue by payment method, cancellation reasons, and ratings comparison between customers and drivers.

The dashboard is designed to give a complete operational overview of ride services and improve business efficiency through data analytics techniques.

2. Dataset Summary

The dataset used in this project contains ride-booking information collected from OLA operations. It includes details related to bookings, customers, vehicle types, ride distances, cancellations, payment methods, ratings, and booking values.

Main Data Fields:

- Booking ID
- Date & Time
- Customer ID
- Vehicle Type
- Pickup & Drop Location
- Booking Status
- Ride Distance
- Booking Value
- Payment Method
- Driver Ratings
- Customer Ratings
- Cancellation Reasons
- Incomplete Ride Reasons

3. Exploratory Data Analysis using Python

We began with data preparation and cleaning in Python:

- **Data Loading:** Imported the dataset using `pandas`.
- **Initial Exploration:** Used `df.info()` to check structure for summary statistics.

```
[2]: import pandas as pd
df = pd.read_excel(r"C:\Users\hp\Downloads\ola_project.xlsx")
df.head()
```

[2]:

	Date	Time	Booking_ID	Booking_Status	Customer_ID	Vehicle_Type	Pickup_Location	Drop_Location	V_TAT	C_TAT	Canceled_Rides_by_Customer	Canceled
0	2024-07-26	14:00:00	CNR7153255142	Canceled by Driver	CID713523	Prime Sedan	Tumkur Road	RT Nagar	NaN	NaN	NaN	Pers
1	2024-07-25	22:20:00	CNR2940424040	Success	CID225428	Bike	Magadi Road	Varthur	203.0	30.0	NaN	
2	2024-07-30	19:59:00	CNR2982357879	Success	CID270156	Prime SUV	Sahakar Nagar	Varthur	238.0	130.0	NaN	
3	2024-07-22	03:15:00	CNR2395710036	Canceled by Customer	CID581320	eBike	HSR Layout	Vijayanagar	NaN	NaN	Driver is not moving towards pickup location	
4	2024-07-02	09:02:00	CNR1797421769	Success	CID939555	Mini	Rajajinagar	Chamarajpet	252.0	80.0	NaN	

- **Missing Data Handling:** Checked for null values and imputed missing values in the `Review Rating` column using the median rating of each product category.
- **Column Standardization:** Renamed columns to **snake case** for better readability and documentation.
- **Database Integration:** Connected Python script to PostgreSQL and loaded the cleaned DataFrame into the database for SQL analysis.

4. Data Analysis using SQL (Business Transactions)

We performed structured analysis in PostgreSQL to answer key business questions:

1. RETRIEVE ALL SUCCESSFUL BOOKINGS

	booking_id text	booking_status text
1	CNR2940424040	Success
2	CNR2982357879	Success
3	CNR1797421769	Success
4	CNR8787177882	Success
5	CNR3612067560	Success
6	CNR4787583516	Success
7	CNR7943634301	Success
8	CNR4524472111	Success
9	CNR8181602032	Success
10	CNR8090918544	Success
11	CNR3196156650	Success
12	CNR9975925287	Success
13	CNR1591113431	Success

2. FIND AVG RIDE DISTANCE FOR EACH VEHICLE

	min_driver_rating_of_sedan double precision	max_driver_rating_of_sedan double precision
1	3	5

3. TOP 5 CUSTOMERS

	customer_id text	total_rides bigint
1	CID268274	4
2	CID635963	4
3	CID836942	4
4	CID266327	4
5	CID288207	4

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Query Query History

```

17 order by total_rides desc limit 5
18
19 --find min and max driver rating for prime sedan bookings
20 select min(driver_ratings) as min_driver_rating_of_sedan,
21        max(driver_ratings) as max_driver_rating_of_sedan
22 from ola where (vehicle_type = 'Prime Sedan')
23
24 --retrieve all rides where payment were made using upi
25 select COUNT(*) booking_id from ola where (payment_method = 'UPI')
26
27 --find average customer rating per vehicle type
28 select vehicle_type, round(avg(customer_rating)::numeric, 2) as avg_customer_rating
29 group by vehicle_type order by avg_customer_rating desc
30
31 --list all incomplete rides with the reason
32 SELECT incomplete_rides_reason,
33        COUNT(*) AS total
34 FROM ola
35 WHERE incomplete_rides = 'Yes'
36 GROUP BY incomplete_rides_reason
37 order by total

```

Data Output Messages Notifications

Showing rows: 1 to 7 Page No: 1 of 1

vehicle_type	avg_customer_rating
Prime Plus	4.01
Auto	4.00
Prime Sedan	4.00
Mini	4.00
Prime SUV	4.00
eBike	3.99
Bike	3.99

Successfully run. Total query runtime: 134 msec. 7 rows affected.

Total rows: 7 Query complete 00:00:00.134

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File Object Tools Edit View Window Help

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36 GROUP BY incomplete_rides_reason
37 order by total

```

Data Output Messages Notifications

Showing rows: 1 to 3 Page No: 1 of 1

incomplete_rides_reason	total
Other Issue	724
Vehicle Breakdown	1001
Customer Demand	1001

Total rows: 3 Query complete 00:00:00.124

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File Object Tools Edit View Window Help

Query Query History

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18
19 --find min and max driver rating for prime sedan bookings
20 select min(driver_ratings) as min_driver_rating_of_sedan,
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Data Output Messages Notifications

Showing rows: 1 to 7 Page No: 1 of 1

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Prime Plus	4.01
Auto	4.00
Prime Sedan	4.00
Mini	4.00
Prime SUV	4.00
eBike	3.99
Bike	3.99

Total rows: 7 Query complete 00:00:00.134

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File Object Tools Edit View Window Help

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Auto	4.00
Prime Sedan	4.00
Mini	4.00
Prime SUV	4.00
eBike	3.99
Bike	3.99

Total rows: 7 Query complete 00:00:00.134

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File Object Tools Edit View Window Help

Query Query History

```

1 SELECT * FROM ola;
2
3 --retrieve all successful bookings
4 select booking_id, booking_status from ola where (booking_status = 'Success')
5
6 --find average ride distance from each vehicle
7 select vehicle_type as vehicle, round(avg(ride_distance), 2)
8 as avg_distance from ola
9 group by vehicle_type
10
11
12
13 --list of top 5 customers who books the highest number of rides
14 select customer_id, count(booking_id) as total_rides
15 from ola
16 group by customer_id
17 order by total_rides desc limit 5
18
19 --find min and max driver rating for prime sedan bookings
20 select min(driver_ratings) as min_driver_rating_of_sedan,
21        max(driver_ratings) as max_driver_rating_of_sedan

```

Data Output Messages Notifications

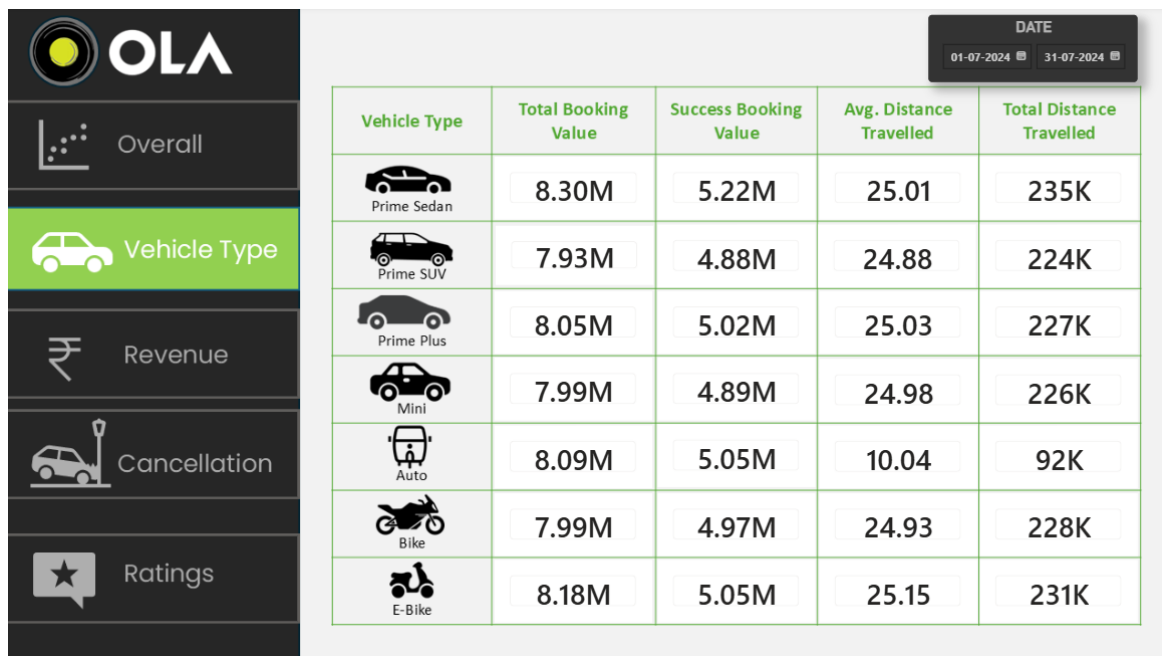
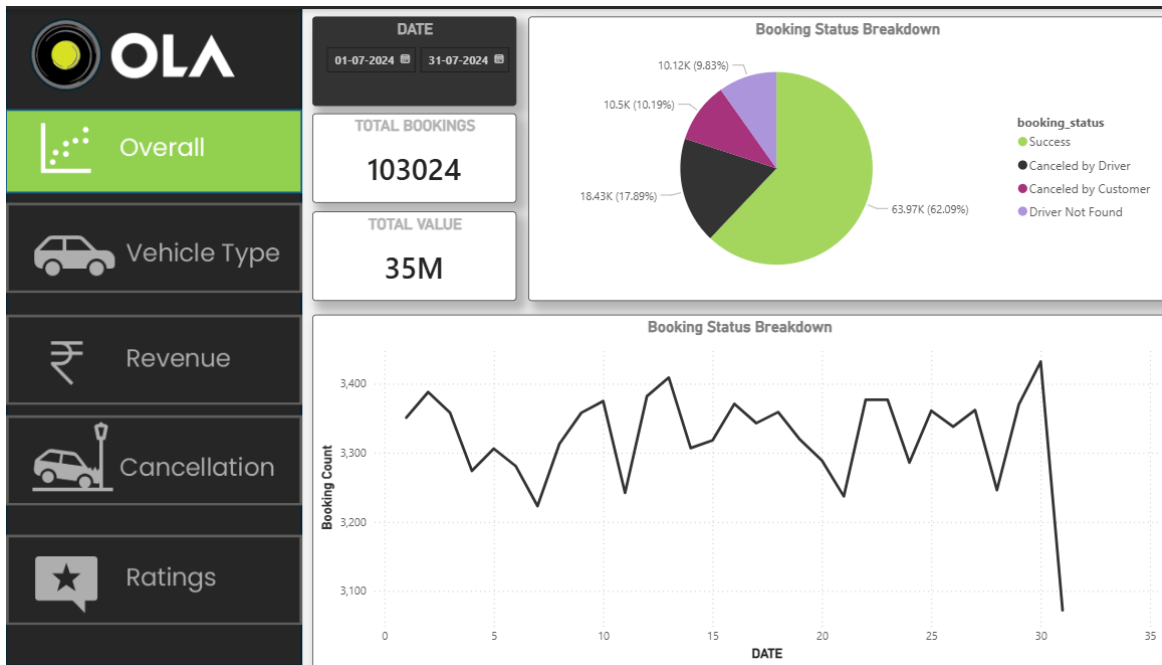
Showing rows: 1 to 1000 Page No: 1 of 64

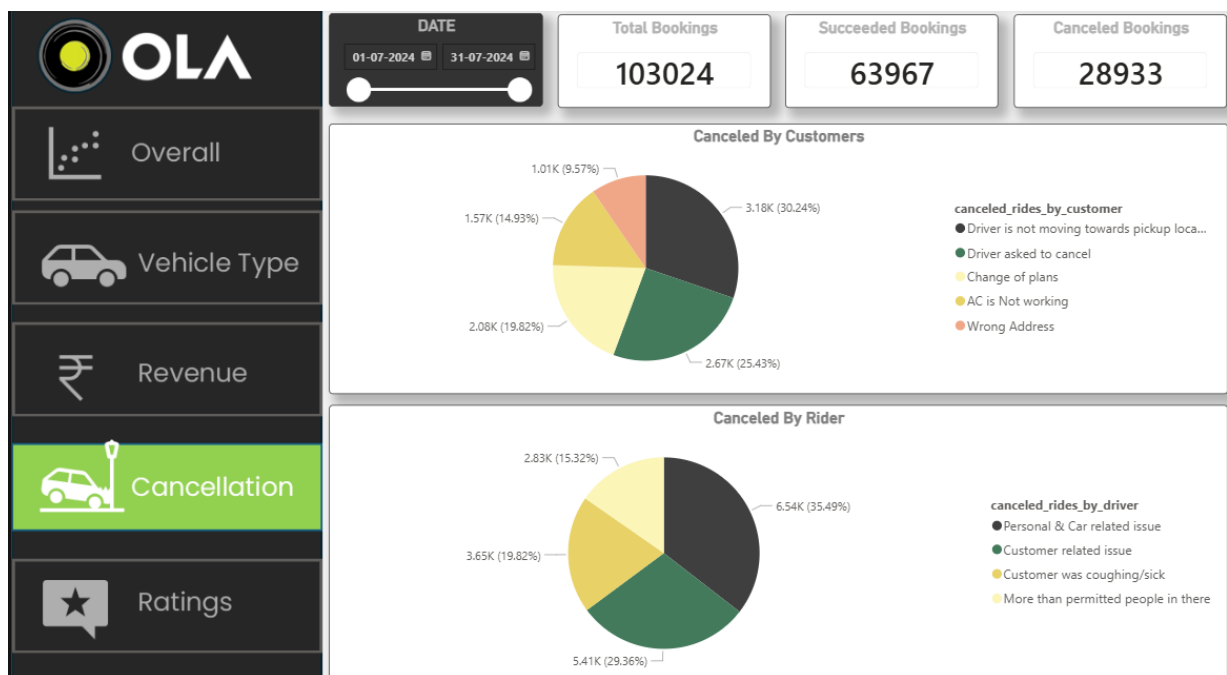
booking_id	booking_status
CNR294324040	Success
CNR294324079	Success
CNR794324079	Success
CNR872777882	Success
CNR3612267860	Success
CNR878785516	Success
CNR794324079	Success
CNR842427111	Success
CNR878785516	Success
CNR8000010244	Success
CNR878785516	Success
CNR878785516	Success
CNR878785516	Success

Total rows: 63967 Query complete 00:00:00.148

5. Dashboard in Power BI

Finally, we built an interactive dashboard in **Power BI** to present insights visually.





Overall

Vehicle Type

Revenue

Cancellation

Ratings

DATE

01-07-2024

31-07-2024

CUSTOMER RATINGS

Prime Sedan	Prime SUV	Prime Plus	Mini	Auto	Bike	E-Bike
4.00	4.00	4.00	4.00	4.00	4.00	3.99

DRIVER RATINGS

Prime Sedan	Prime SUV	Prime Plus	Mini	Auto	Bike	E-Bike
3.99	4.00	4.00	4.00	4.00	3.99	4.00